0580_w24_ms_42_Q8

| Question | Answer | Marks | Partial Marks | Guidance |
|---|---|---|---|---|
| 8(a)(i) | $\frac{5}{12}$ oe | **1** | | In all parts:<br>Only accept fraction, percentage and decimal equivalents (3sf or better)<br>Ignore incorrect cancelling or conversion following a correct answer.<br>Penalty –1 once for 2sf or words e.g. 1 in 2 or 1 out of 2<br>**But not** ratio, not 1 to 2<br><br>0.417 or 0.4166 to 0.4167 |
| 8(a)(ii) | $\frac{7}{12}$ oe | **1** | **FT** 1 – *their* (a)(i) | 0 < *their* (a)(i) < 1<br>0.583 or 0.5833… |
| 8(b)(i) | $\frac{4}{35}$ oe | **2** | **M1** for $\frac{4}{15} \times \frac{6}{14}$ | 0.114 or 0.1142 to 0.1143<br>M0 for $2 \times \frac{4}{15} \times \frac{6}{14}$ |

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|---|---|---|---|---|
| 8(b)(ii) | $\frac{74}{105}$ oe | 4 | **M3** for $1-\left(\frac{5}{15}\times\frac{4}{14}+\frac{6}{15}\times\frac{5}{14}+\frac{4}{15}\times\frac{3}{14}\right)$ oe<br>or **M2** for $\frac{5}{15}\times\frac{4}{14}+\frac{6}{15}\times\frac{5}{14}+\frac{4}{15}\times\frac{3}{14}$ oe<br>or **M1** for $\frac{k}{15}\times\frac{k-1}{14}$ where $k$ is 4, 5 or 6 oe<br>If 0 scored, **SC1** for $\frac{148}{225}$<br><br>ALTERNATIVE 1<br>**M3** for $\frac{5}{15}\times\frac{10}{14}+\frac{6}{15}\times\frac{9}{14}+\frac{4}{15}\times\frac{11}{14}$ oe<br>or **M2** for two of these products added oe<br>or **M1** for $\frac{k}{15}\times\frac{15-k}{14}$ where $k$ is 4, 5 or 6 oe<br>If 0 scored, **SC1** for $\frac{148}{225}$<br><br>ALTERNATIVE 2<br>**M3** for $\frac{5}{15}\times\frac{6}{14}\times 2+\frac{5}{15}\times\frac{4}{14}\times 2+\frac{6}{15}\times\frac{4}{14}\times 2$ oe<br>or **M2** for at least two of these different products added oe<br>or **M1** for one correct product<br>If 0 scored, **SC1** for $\frac{148}{225}$ | 0.705 or 0.7047 to 0.7048<br>1 – (P(GG) + P(WW) + P(BB))<br>For all method marks, products must be shown<br><br>For M1 isw incorrect products added<br><br>P(GG') + P(WW') + P(BB')<br><br>For M1 isw additional incorrect products added<br><br>P(GW) + P(GB) + P(WB)<br>For M2 e.g. $\frac{5}{15}\times\frac{6}{14}+\frac{5}{15}\times\frac{4}{14}$ oe<br><br>For M1 isw additional incorrect products added |